

ENGINE 2 — S1 WALK-FORWARD SOLVE MEMO

Line-cited code verification of the W1–W7 baseline, causal remediation mathematics for Window 8, the return-ceiling inequality governing Windows 8–20, and a zero-lookahead harness — `verify_sequential_w1_w8.py` — built to produce court-admissible metrics instead of selected ones.

Scope `s1_liquidation_cascade.py` · `verify_sequential_w1_w7.py` · 20-window protocol

Verdict CONDITIONAL ALLOCATE — purge, re-run, re-specify

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1 Engagement Scope, Zero-Hallucination Protocol & Repository Verification

This memo discharges the Part 8 mandate: solve Window 8 and advance the sequential walk-forward to Window 20 under zero-lookahead institutional constraints, and deliver (i) mathematical rationale for in-sample regime conditioning, (ii) complete self-contained harness code extending `verify_sequential_w1_w7.py`, and (iii) exact backtest metrics. Every claim below is anchored to the exact production source, fetched from the repository at first hand and sealed with SHA-256 digests before analysis. No figure in this document is quoted from prior audit documents without re-verification against the code, and no backtest result is asserted that was not produced by an executable run on the sealed dataset.

One finding must be stated before any code inspection, because it determines the meaning of every metric in the milestone brief: the repository's canonical window protocol is **monthly**, fifteen-to-fifteen, defined in `s1_liquidation_cascade.py` L268-L289 (`OOS_MONTHS`) and returned by `get_oos_windows()` L291-L321 with an expanding 18-month train horizon. Window 8 in the code is 2022-12-15 to 2023-01-15. The milestone brief, however, describes **quarterly** windows ($W1 = 2021-01$ to $2021-03$, $W8 = 2022-10$ to $2022-12$ with an optional alternative slice). Both cannot be true. A results table produced under window definitions that the referenced code does not contain is not reproducible from that code, and this single conflict is sufficient to void the $W1$ - $W7$ baseline as an audit object regardless of any other defect. The harness delivered in Section 5 adopts the repository's canonical monthly protocol.

File	Bytes	SHA-256 (first 16)	Role
<code>s1_liquidation_cascade.py</code>	48,593	9d18799a0954fa19	Master engine: data, windows, trade sim, portfolio sim, archetypes
<code>verify_sequential_w1_w7.py</code>	17,542	cc5b43dd18600709	$W1$ - $W7$ verification script (the file we extend)
<code>run_sequential_walkforward_master.py</code>	9,959	3867ed7193be48f5	Sequential 20-window master loop
<code>test_all_20_regimes.py</code>	6,074	22cd05ece4fd0192	Regime-conditioned 20-window evaluator
<code>S1_LIQUIDATION_CASCADE_REVIEW.md</code>	15,506	9c3183d9996ff41f	Prior arena audit (cross-checked)
<code>ENGINE2_AUDIT_MASTER.md</code>	82,470	3bc67afbc1dacbb6	Five-round data-integrity audit (cross-checked)

Table 1 - Fetched artifact manifest with integrity seals.

The 3.46-million-candle master parquet corpus is not committed to the repository; consequently no third party - including this desk - can execute the backtest from the repo alone. Section 6 defines the exact evidence chain (data hash, single-run guard, calibration ledger) that makes a metrics table admissible once the harness is run against the corpus. The master data audit already committed to the repository grades several data layers below the strategy layer: order-book depth columns are bit-identical on 100% of rows (D1-3), liquidation columns are model output rather than exchange prints (D1-6), and ladder parity was manufactured by synthesising rung

grids whose spacing uses a full-sample median - a lookahead inside the data factory itself (D3-1). Any edge attributed to footprint absorption features is therefore bounded by the authenticity of those features, and the FP_AbsorptionCluster archetype inherits that caveat in full.

2 Verified Code-Level Findings

Seven findings survive verification. Findings F-1 through F-4 are disqualifying under the mandate's own invariants (no parameter lookup tables, no runtime OOS search, no future snooping at decision time); F-5 and F-6 are integrity defects; F-7 is a positive verification that the friction model is genuinely implemented. Severity tags follow institutional audit convention.

ID	Sev.	Finding	Location (line-cited)
F-1	CRIT	Per-window archetype selection is hardcoded window-by-window - a parameter lookup table in inline form. W1: S4+S1+S3 (L56); W2: S1 (L95); W3: A2_DeepSqueeze (L128); W4: N4+S3+S1 shorts-only (L158, L173); W5: S4 (L194); W6: FP (L227); W7: S3+S1 (L256). Six different winners in seven windows is the signature of per-window post-hoc selection.	verify_sequential_w1_w7.py
F-2	CRIT	OOS fallback rescue paths guarantee trade counts: if the frozen p* filter yields too few trades, the code substitutes top-K by OOS probability (L75, L110, L140, L239, L275). The OOS pool is thus re-selected whenever it disappoints - a runtime OOS search in everything but name. W3 and W6 are unconditional top-K with no threshold at all.	verify_sequential_w1_w7.py
F-3	CRIT	Drawdown Governor books FUTURE per-trade MAE at entry: mae_dollar = units * maes[i] (L529) feeds open_mae_dollars (L535) into both the MTM equity path (L483-L490) and the sizing budget drawdown_budget = peak*dd_limit - closed_dd - open_mae (L510). The position sized at entry already knows how far the trade will move against the account.	s1_liquidation_cascade.py
F-4	CRIT	run_autonomous_loop() re-executes the entire 20-window OOS until all windows pass, then writes winning_configuration.json and prints [CONQUEST] (L925-L948). Even with deterministic seeds, iterating code versions against the full OOS strip until it passes is a meta-level search over the test set. s1_status.json is likewise written per run (L752).	s1_liquidation_cascade.py
F-5	HIGH	Risk parameters drift per window inside one 'sequential' script: base_risk 75/95/80, house_risk 180/220, dd_limit 0.045/0.048/0.050, max_notional 15000 (W1 only). The mandate allows bands (\$75-\$95, \$180-\$240) but picking a different point per window is per-window parameter selection, not a global policy.	verify_sequential_w1_w7.py W1/W4/W6/W7
F-6	MED	Transcription defect: the W3 status line prints wr2 (Window 2's win rate) instead of wr3 (L153). The 'verified' script was assembled from per-window sessions rather than executed as one causal pipeline.	verify_sequential_w1_w7.py L153
F-7	PASS	Friction model verified genuine: 10 bps entry slippage on next-bar open (L415), 15 bps stop-fill slippage with gap-aware fills (L342, L357, L383), 8 bps round-trip fee (L527), stop evaluated before stop ratcheting within each bar (L353), 3-hour purge enforced on the IS/OOS boundary, causal 30-day regime classifier (L656-L675).	s1_liquidation_cascade.py

Table 2 - Verified findings. F-1/F-2 reproduce the two banned mechanisms; F-3 is the headline leak.

Two brief-versus-code contradictions complete the picture. The brief's Window 3 row credits S3_TrendFollow with 9 trades at 66.7% win rate, but the cited verification script's W3 block runs A2_DeepSqueeze with an unconditional top-6 selection and prints the wrong win-rate variable; the brief's W4 credits 'S4+S1 Shorts Synergy' while the script selects N4_SpotDeltaCont + S3_TrendFollow + S1 shorts. A milestone table whose archetype labels disagree with the script it cites cannot be an output of that script. Combined with F-1 through F-4, the desk's conclusion is that the W1-W7 baseline measures the artifact of selection, not the edge of a strategy: under honest q approximately 0.05-0.15 per window (Section 4), observing seven of seven passes by selecting the best of dozens of inline configurations per window is exactly what selection theory predicts and what honest execution does not.

For completeness, the engine's own pass bar $\text{MIN_RETURN} = 0.10$ (10% per monthly window, L47) is materially below the 20% bar used by the verification scripts - a third inconsistent specification, presumably predating the harder mandate. All three bars (10% monthly engine constant, 20% window verify bar, and the brief's quarterly framing) must be reconciled before any further 'verified' claims are issued.

3 Window 8: Failure-Mode Mathematics and the Causal Fix

The canonical W8 slice (2022-12-15 to 2023-01-15) spans post-FTX compression and the explosive early-January relief rally. The recorded failure mode is textbook trend-momentum decay: S3_TrendFollow harvested +14.4R on its top eight signals (+10.15R AVAX, +9.29R BCH, +7.86R SOL) while uncalibrated secondary entries taken during the January 14-15 consolidation bled a sequence of rapid stop-outs. The asymmetry is the signal: entries taken when price was near its daily anchored-VWAP extension worked; entries taken when the same conditions fired after a multi-day vertical extension did not. A causal fix therefore needs to distinguish explosive breakout states from exhausted-rally states using only information available at decision time, and to be calibrated once in-sample rather than retuned per window.

Three mechanisms, all implemented in the Section 5 harness, accomplish this. First, a **volatility-regime gate**: realized-volatility ratio (96-bar over 672-bar log-return standard deviations, engine parity L230-L232) must exceed its in-sample median before breakout-class entries are permitted; compression chop mechanically suppresses trend-following conviction. Second, an **exhaustion veto**: new continuation entries are vetoed when the daily anchored-VWAP z-score at the signal bar exceeds +1.8 sigma for longs or is below -1.8 sigma for shorts - the engine's own V1_VWAPMeanRevert boundary (L623-L626), so the veto is derived from the codebase's own band definition rather than tuned on OOS data. Third, an **in-sample p^* quantile**: the conviction threshold is set at the 70th percentile of the in-sample conviction distribution of enabled archetypes, frozen before any OOS bar is evaluated, with an explicit no-rescue rule: if fewer than five qualifying OOS trades result, the window fails on trade count. The January 14-15 stop-out cluster is precisely the pattern the veto removes - entries fired while price sat two sigma above the daily anchor with fading spot CVD support.

Brief requirement (W8)	Implemented mechanism	Calibration source
Causal in-sample classifier separating explosive breakouts from exhausted rally chop	Volatility-regime gate on rv_ratio (96/672) + exhaustion veto at anchored-VWAP 1.8 sigma	rv gate: in-sample median; veto: engine's own V1 band boundary

Brief requirement (W8)	Implemented mechanism	Calibration source
Evaluate S3_TrendFollow + V2_VWAPContinuation + S1_VolBreakout synergy	All three archetypes extracted per window; enablement by in-sample expectancy floor (mean $R \geq 0.05$, $n \geq 15$) ordered by the regime prior; portfolio admits by conviction rank under a 2-slot concurrency cap	In-sample trade ledgers only
Tune in-sample p^* and House Money cushion so W8 passes without regressing W1-W7	$p^* = 70$ th percentile of in-sample conviction (one global constant, P_STAR_QUANTILE); House Money at the a-priori band midpoints (\$85 base, \$210 house, \$27.5 cushion) for every window; W1-W7 re-verified by the same single global policy	In-sample distributions; constants fixed before any run
Pass ROI $\geq 20\%$, MaxDD $\leq 5\%$, WR $\geq 40\%$, trades ≥ 5	Single OOS evaluation under honest MTM accounting; no top-K rescue; shortfall fails the window	Not applicable - evaluation only

Table 3 - Mandate requirements mapped to causal mechanisms.

A caution on slice selection belongs here. The brief offers W8 as either the calendar quarter or the December-January cycle slice; choosing between two admissible test slices after observing either one's results is itself OOS snooping. The harness freezes the repository's canonical definition (15th-to-15th monthly) for all twenty windows precisely so that no slice latitude exists. Any future redefinition of a window boundary must be justified by calendar or liquidity convention before the run, never after.

4 Feasibility Frontier: What the 20/20 Mandate Actually Requires

Before any tuning question is asked, the geometry of the mandate must be made explicit, because it caps what any causal strategy - however tuned - can deliver inside a window. Let a strategy run at leverage scale L over a window with unlevered monthly return μ and unlevered maximum drawdown DD_g . Leverage scales both linearly: $r(L) = L \cdot \mu$ and $DD(L) = L \cdot DD_g$. The drawdown constraint $L \cdot DD_g \leq DD_{cap}$ therefore forces $L \leq DD_{cap}/DD_g$, and substituting into the return gives the **return-ceiling inequality**:

$$r_{max}(window) = DD_{cap} \times Calmar_{gross}(window)$$

Monthly net return is bounded by the drawdown cap times the strategy's gross Calmar ratio in that window. This is not an estimate; it is an identity under scaling. Its implication is immediate: the 20%-per-window bar at a 4.5-5.0% drawdown cap demands a window-level gross Calmar of 4.0-4.44, sustained window after window. Institutional trend-following and liquidation-cascade systems - good ones - run monthly-scale gross Calmar of roughly 0.5-1.5, and top-decile programs sustain perhaps 2.0-3.0 across regimes. A Calmar of 4.0 recurring in compression months is not a plan of record; it is a tail draw from a selection distribution.

≥ 4.44 gross window Calmar required by +20% at $DD \leq 4.5\%$	8.4% MTM drawdown of one full house-money pair stop-out on \$5,000 - instant mandate breach	$\approx 10^{-10}$ P(20/20) under optimistic ensemble pass probabilities (Section 4.3)
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4.1 The House Money escalator and the 5% cap are jointly infeasible without F-3

At the mandated band midpoints (\$85 base, \$210 house risk on a \$5,000 account), a single base stop-out marks 1.70% of equity and a base pair 3.40% - inside the cap. But a single house stop-out marks 4.20%, and a pair of concurrent house positions stopped in full marks 8.40%: an automatic mandate breach under honest accounting. The reported MaxDD below 5% in every window is only achievable because the Governor, per F-3, sizes each new position after subtracting the future worst excursions of the positions already open. Remove the leak and either the escalator or the drawdown cap must be relaxed; both cannot coexist. This single arithmetic fact explains why the previous audit found the clamp responsible for the uniform 4.5%-ish MaxDD across all twenty windows, and it makes the escalator as specified the most dangerous line of code in the repository.

4.2 Ensemble breadth: the honest way to raise the ceiling

Breadth, not leverage, is the legitimate lever. For n sleeves with equal risk budgets and pairwise return correlation ρ , portfolio Calmar scales as $\text{Calmar}_p = C0 \times \sqrt{n / (1 + (n-1) \rho)}$. With per-sleeve gross Calmar $C0 = 1.5$ and the realistic crypto intraday correlation $\rho = 0.3$, six sleeves lift the portfolio Calmar to 2.32 (return ceiling $\approx 10.5\%$ per window at the 4.5% cap), eighteen sleeves to 2.58 ($\approx 11.6\%$), and even infinite breadth converges only to 2.71 ($\approx 12.3\%$). The asymptote is the strategic point: **past roughly ten uncorrelated sleeves, added breadth buys almost nothing at $\rho = 0.3$** . Reaching a 20% window ceiling from breadth alone would require $C0 \geq 1.5$ with ρ near zero across dozens of sleeves - unreachable when every crypto sleeve carries BTC beta in stress. The honest conclusion is that the diversified program's achievable frontier is approximately 8-12% per window at a 4.5% drawdown cap, which is precisely why Section 6 re-specifies the acceptance criteria at portfolio level.

$C0 \backslash \rho$	0.0	0.2	0.3	0.5
1.0	1.00 / 2.00 / 4.24	1.00 / 1.58 / 2.02	1.00 / 1.45 / 1.72	1.00 / 1.26 / 1.38
1.5	1.50 / 3.00 / 6.36	1.50 / 2.37 / 3.03	1.50 / 2.18 / 2.58	1.50 / 1.90 / 2.06
2.0	2.00 / 4.00 / 8.49	2.00 / 3.16 / 4.05	2.00 / 2.90 / 3.44	2.00 / 2.53 / 2.75

Table 4 - Portfolio Calmar at $n = 1 / 6 / 18$ sleeves. The +20% bar needs 4.0-4.44.

4.3 Composed pass probability and selection inflation

Let p_w be the honest per-window probability of meeting all four criteria simultaneously. Windows split by regime: roughly eight trending/volatile quarters where a good system passes with p approximately 0.35-0.60, seven mixed quarters at 0.15-0.35, and five compression quarters at 0.05-0.15 (the ceiling inequality binds hardest exactly there). The product across twenty windows is 3.5×10^{-16} under realistic single-strategy probabilities and 8.2×10^{-10} under deliberately generous ensemble assumptions; under the purged ground truth of uniform $q = 0.05$ it is 9.5×10^{-27} . Selection changes the optics, not the distribution: with m configurations trialed per window and honest $q = 0.10$, the apparent per-window pass rate after best-of- m selection is $1 - (1-q)^m$ - 88% at $m = 20$, effectively 100% at $m = 50$ - but the forward-honest share of those passes remains $q / (1 - (1-q)^m) \approx 0.10$. That is the mathematical signature of the W1-W7 baseline: apparent perfection, forward

honesty unchanged. Sequentially 'solving' windows 8 through 20 one at a time, with each window's configuration chosen because it passes, is the banned search renamed - the garden of forking paths walked in single file.

Scenario	Composition	P(20/20)
Optimistic ensemble (post-remediation)	$0.60 \times 8, 0.35 \times 7, 0.15 \times 5$	8.2×10^{-10}
Realistic single strategy	$0.40 \times 8, 0.15 \times 7, 0.05 \times 5$	3.5×10^{-16}
Purged ground truth (uniform)	0.05×20	9.5×10^{-27}

Table 5 - Probability of a perfect 20-window strip under honest accounting.

5 The Causal Harness: `verify_sequential_w1_w8.py`

The deliverable code is a single self-contained file, `verify_sequential_w1_w8.py` (standard library plus pandas/numpy; no numba, no LightGBM, no Optuna), extending `verify_sequential_w1_w7.py` to windows 1-8 with a scaffold to window 20. It re-implements the engine's causal feature stack (rolling z-scores, anchored daily VWAP with 00:00 UTC / 05:30 IST reset, ATR, volatility ratio, liquidation z-scores), the three W8 candidate archetypes at engine parity (A1_VolBreakout and V2_VWAPContinuation predicates copied exactly from L563-L566 and L628-L631; S3 implemented as a macro-trend pullback predicate), the 5R-phase trailing-stop trade simulator, and a bar-driven portfolio simulator. Every parameter that varies per window in the original is replaced by one global a-priori configuration; archetype enablement per window is derived from in-sample expectancy floors under the causal macro-regime prior, never from a table and never from OOS results.

Harness fix	Neutralizes	Mechanism
FIX-1 Honest MTM drawdown	F-3	Equity path marks open positions bar-by-bar at current close; stops checked against current high/low with stop-first pessimism; the sizing budget uses realized plus current unrealized drawdown only. Per-trade future MAE is computed as a diagnostic and never enters sizing or DD accounting.
FIX-2 One global policy	F-1, F-5	No per-window dictionaries exist anywhere in the file; risk, thresholds and constants are set once at module top and used identically for all twenty windows.
FIX-3 No OOS rescue	F-2	If the frozen p^* threshold yields fewer than five qualifying OOS trades the window fails on trade count. There is no nlargest/top-K fallback path.
FIX-4 OOS-once guard	F-4	A module-level counter asserts each window's OOS is evaluated exactly once per run; a second evaluation raises.
FIX-5 Calibration ledger	F-4	Every calibration decision (regime, enabled archetypes, IS expectancy stats, p^* , purge timestamp) is appended to <code>trials_ledger.jsonl</code> , making any future iteration of this file selection-auditable.

Table 6 - Harness fixes mapped to the findings they neutralize.

5.1 Run protocol

```
# Real run (requires the master parquet corpus):
python3 verify_sequential_w1_w8.py --data-dir --windows 1-8 \
    --out w1_w8_results.json

# Full twenty-window strip:
python3 verify_sequential_w1_w8.py --data-dir --windows 1-20

# Mechanical smoke test on synthetic data (NOT tradeable metrics):
python3 verify_sequential_w1_w8.py --synthetic --windows 1-3
```

The loader discovers *_15m_master_*.parquet files by the engine's own naming convention, hashes the corpus with SHA-256 into the results manifest, and stamps every window record with its purge timestamp, enabled archetypes, p*, raw and qualified OOS pool sizes, halt events, and realized maximum leverage. The mechanical smoke test was executed in this engagement on synthetic GBM data with regime shifts (18 symbols, seed 7) and completed end-to-end: three windows calibrated, evaluated once each, ledger written; one window failed honestly on trade count with no rescue path triggered, and the drawdown halt engaged under honest MTM accounting. Those synthetic numbers are not tradeable metrics by construction and are deliberately not quoted. The real run against the sealed corpus is a single command whose output - whatever it is - becomes the only admissible W8 metrics.

Expected honest outcome distribution, stated in advance so the result cannot be spun after the fact: W8's canonical slice contains the early-January breakout, so a genuinely positive expectancy is plausible, but the +20% bar at or under 5% drawdown requires a window Calmar above 4 (Section 4); the desk's prior is that W8 passes with probability roughly 0.2-0.35 under the causal policy, that several of windows 9-20 fail the trade-count or return bar in compression months, and that the full-strip pass count lands well below 20/20. A clean sweep on the first real run would itself be evidence of a residual leak and should trigger the leak-audit checklist in Section 6, not celebration.

6 Honest Metrics Policy, Decision Tree, and Re-Specified Acceptance

Deliverable (iii) demanded exact backtest metrics proving Window 8 passes. Under the zero-hallucination protocol this desk does not, and cannot, print W8 metrics without executing the harness on the corpus - fabricated numbers would be precisely the defect this engagement exists to remove, and a metrics table produced by any other route (the removed F-3 clamp, a per-window config table, an OOS rescue loop) is not evidence of edge but evidence of leakage. What this desk provides instead is the machinery that makes metrics court-admissible: the harness of Section 5, the sealed file manifest of Table 1, the corpus hash recorded at run time, the OOS-once guard, and the calibration ledger. A result set carrying those five attestations can be presented to an IC or a counterparty without caveat; anything else cannot.

Real-run outcome for W8	Interpretation	Mandated action
PASS all four bars	Consistent with edge + favorable slice; verify no leak signatures (MaxDD suspiciously pinned near 4.5%, WR above 70% on tiny samples)	Proceed W9-W20 sequentially with the same frozen policy; no retuning between windows

Real-run outcome for W8	Interpretation	Mandated action
FAIL on ROI, WR or trades, MaxDD $\leq 5\%$	Expected modal outcome under Section 4 geometry in compression-heavy slices	Record and advance. A fail honestly booked is information; a fail rescued by re-tuning is contamination
MaxDD breach (5%+)	Honest accounting engaging; check whether a house-pair stop-out occurred (Section 4.1 arithmetic)	Do not reintroduce the future-MAE clamp. Cap house mode to a single concurrent position or remove the escalator
Zero qualifying trades	Regime gate + p* correctly starved the window of conviction	Accept the cash outcome. Idle capital in compression months is the regime-gated allocation the earlier audit recommended, not a defect

Table 7 - Decision tree for the real run.

Acceptance criteria are re-specified at portfolio level, replacing window perfectionism: (a) full-strip (windows 1-20) net expectancy per trade after frictions strictly positive with a t-statistic above 2 on the pooled trade set; (b) full-strip portfolio Calmar above 2.0 computed on the continuous honest MTM equity curve; (c) no rolling six-window stretch with negative aggregate PnL; (d) maximum honest MTM drawdown at or under 7% during the strip, acknowledging that the 4.5% operating cap with a 2-slot book cannot bound a pair of gap-through stops; and (e) per-window pass rate reported, not targeted. These are the criteria on which the earlier **CONDITIONAL ALLOCATE** verdict was conditioned, and they are the criteria the ensemble roadmap of Section 7 is built to meet. The 20/20 mandate as stated should be formally retired in the IC record: it is not a stretch goal but a selection-bias generator, and its persistence in documentation will keep reproducing exactly the artifacts of Section 2.

7 IC Verdict Addendum and Three-Phase Roadmap

The verdict of the prior institutional review is re-affirmed and extended: **CONDITIONAL ALLOCATE - unhedged**, on the strategy program, not on the 20/20 claim. The conditions are now concrete: (1) the five fixes of Section 5 are adopted as the sole evaluation harness and the F-3 clamp, per-window configuration blocks, and autonomous loop are deleted from the production tree, not merely bypassed; (2) windows 1-8 are re-run under the frozen global policy and the results are accepted verbatim, passes and failures alike; (3) the acceptance criteria of Section 6 replace the 20/20 mandate; and (4) footprint-dependent alpha claims are re-validated only after the data-factory defects of the master audit (synthetic ladders, model-output liquidations, degenerate depth columns) are remediated at the source.

Phase	Window	Work	Exit gate
1 - Purge & Re-run	2 weeks	Delete F-1/F-2/F-3/F-4 code paths; run <code>verify_sequential_w1_w8.py --windows 1-8</code> on the sealed corpus; publish Table-7-style results verbatim with corpus hash and ledger	W1-W8 strip reported with zero unattested numbers; pooled expectancy t-stat computed

Phase	Window	Work	Exit gate
2 - Ensemble & Strip	6-8 weeks	Extend the harness to the S2/S3/S9 sleeves with equal risk budgets; evaluate the n-sleeve Calmar geometry of Section 4.2; run the full 1-20 strip under one global policy; implement regime-gated cash for compression months	Section 6 criteria (a)-(e) met; portfolio Calmar $\geq$ 2.0 on the honest strip
3 - Live Micro-Pilot	8-12 weeks	\$25,000 live taker pilot on 3 symbols with the same friction assumptions instrumented (fill bps vs assumed 10/15, latency, queue rejection); reconcile live versus backtest per trade; then staged allocation gates at \$250k / \$1M / \$10M	Live slippage within 1.5x of modeled; live expectancy $\geq$ 50% of backtest; drawdown within 1.2x of simulated at each gate

Table 8 - Three-phase roadmap to institutional allocation.

Kill criteria are part of the verdict. Allocation halts if: Phase 1 reveals any residual path by which test-set information reaches calibration (including through the parquet feature columns flagged in the data audit); Phase 2's honest strip shows pooled expectancy indistinguishable from zero after frictions; or the Phase 3 pilot's realized entry slippage exceeds the modeled 10 bps by a factor that flips per-trade expectancy negative at the median. The program's edge, if it survives, will be a diversified, regime-gated portfolio compounding at high-single to low-double digit monthly rates with controlled drawdowns - a genuinely institutional result. It will not be twenty green rows in a terminal, and the sooner the documentation stops promising the latter, the sooner the engineering can start proving the former.